# Proficiency Test III

Solutions

## Instructions

Put your name at the top of this document in the YAML. You are asked to clone the `proficiency-test-III` repo from GitHub. This repo has been made for you.

If you can not clone the repo, please still create a .qmd file and complete the tasks for partial credit.

If you can not get your code the run, please use the appropriate code chunks to stop your code from evaluating (but still make sure we can see your code) to earn partial credit.

> **Note**
>
> Part of your grade comes from demonstrating proper use of version control.
> You must have 3-4 meaningful commits to earn full credit.
> Each commit should be after you answer a specific task.

Please answer each task below. **Render after each question**

## Task-0 (5-points)

Change the README of your GitHub repo to have a level-2 header that says **mtcars**. Underneath, copy the description of the data set from the help file. Next, put a level-3 header that says **source**, and copy the source from the help file into your README. Commit these changes directly to the main branch so the updates appear on your GitHub repo.

**You can change the README in two different ways. You can either open the README file in RStudio + make edits, or you can change the README directly on GitHub by clicking the pencil and committing to the main branch**

## Task-packages (5-points)

Include a code-chunk that libraries the `tidyverse` package at the top of your document. In this code chunk, turn off all messages and warnings that result from opening `tidyverse`.

## Packages

Don't forget to put the following at the top of your code chunk!

```
#| message: false
#| warning: false
```

```
library(tidyverse)
```

## Task-1 (5-points)

Suppose you are trying to answer the following research question: Using the mtcars dataset, which car, with at least 6 cylinders, has the highest fuel efficiency (mpg)? Report the single row in the data set that answers this question.

A chat bot spits out the following code.

```
mtcars |>
  filter(cyl > 6, mpg > 20) |>
  select(mpg, cyl) |>
  arrange(mpg) |>
  slice_head(n = 5)
```

In your .qmd file, write the correct code in your code chunk that fixes/changes the code chunk above to answer the question.

*Fix below. A lot of unnecessary code that does not pertain to the research question. There is not reason to filter mpg > 20, and nothing in the problem tells us to select only 2 columns. slice_head() is also used incorrectly, and should be set to 1 instead of 5.*

```
mtcars |>
  filter(cyl >= 6) |>
  arrange(mpg) |>
  slice_head(n = 1)
```

```
                  mpg cyl disp  hp drat   wt  qsec vs am gear carb
Cadillac Fleetwood 10.4   8  472 205 2.93 5.25 17.98  0  0    3    4
```

## Task-2 (5-points)

In 2-3 sentences, explain why this code won't work and instead returns an error. Note: You do not have to fix the code, just articulate **why** this code will not run.

```
mtcars |>
  select(mpg, cyl) |>
  filter(cyl >= 6, hp > 150) |>
  arrange(desc(hp)) |>
  slice_head(n = 5)
```

*The code will not work because we are using `filter()` on a column that no longer exsists in our mtcars data set. Order matters, and we are piping a data set with only mpg and cyl into filter.*

## Task-3 (5-points)

Using the `mtcars` dataset, report how many cars have at least 4 cylinders, have a mpg above 15, and have a horsepower **per** cylinder (horsepower divided by cylinder) lower than the mean horsepower **per** cylinder across the 32 cars. Provide the appropriate code in a code-chunk that answers this question.

```
mtcars |>
  mutate(hp_per_cyl = hp / cyl) |>
  filter(cyl >= 4, mpg > 15, hp_per_cyl < mean(hp_per_cyl)) |>
  select(mpg, cyl, hp, hp_per_cyl) |>
  count()
```

## Task-4 (5-points)

The following code below applies a weighted moving average using weights of 0.25, 0.5, and 0.25 using the `filter` function.

For each observation, it calculates a weighted average of the observation and its two neighbors. For example, for the middle observation (6):

$0.25(4) + 0.5(6) + 0.25(8) = 6$

The result is a smoothed version of the original data. Values at the boundaries are NA because there aren't enough neighboring observations to calculate the weighted average.

The function needs to be called from the `stats` package. Type the code below into your .qmd file, and then add to the code below to make the code function as intended. Note, do **not** library the stats package to answer this question.

Reminder: If you can not get this code to work, your document will not render unless you stop the code chunk from evaluating.

*Package masking is when two different packages share a common name. Because we `library(tidyverse)` at the top of the document, the `filter()` function comes from tidyverse, and not the stats package. We can specify a package by the package name and :: next to the function name, as seen below.*

```
stats::filter(
  c(2, 4, 6, 8, 10),
  c(0.25, 0.5, 0.25),
  sides = 2
)
```